MAE C163A/C263A
Kinematics of Robotic Systems

Homogeneous Transformation Matrix

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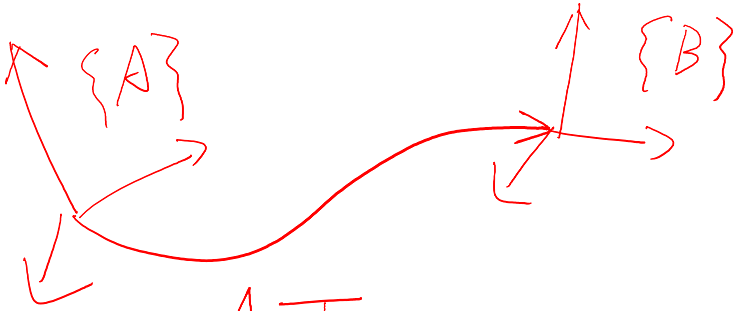
Homogeneous Transformation Matrix

orientation & position



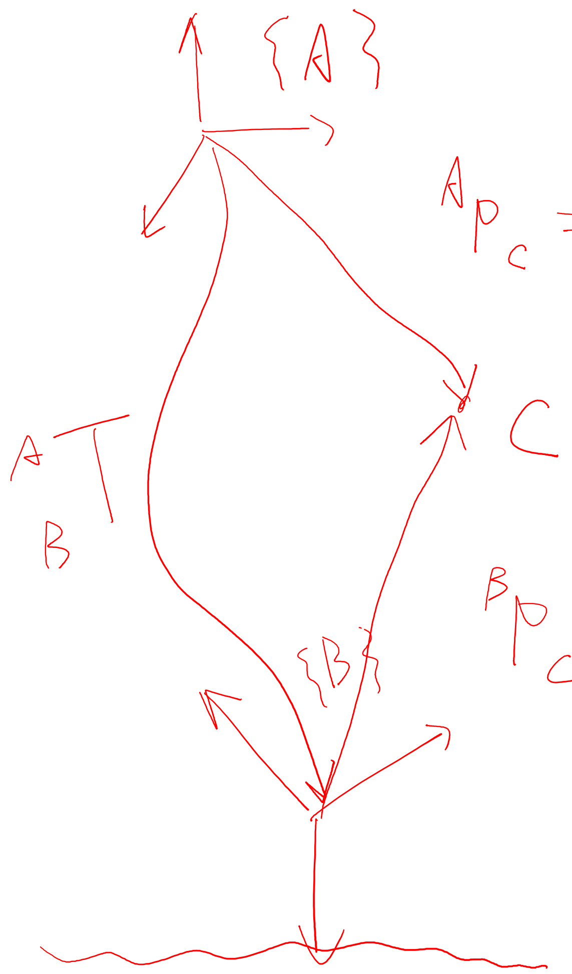
${}^{A}_{B}T$

4x4 matrix

$$\begin{array}{cc} & \begin{array}{cc} 3 & 1 \end{array} \\ \begin{array}{c} 3 \\ 1 \end{array} & \begin{bmatrix} R & p \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} 3 \\ 1 \end{array}$$

special Euclidean group SE(3)

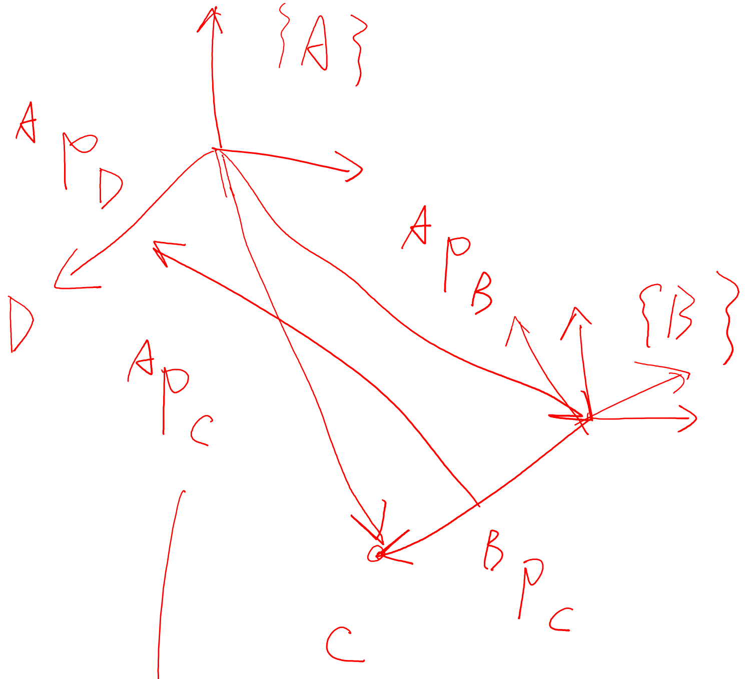
Position Vector



$${}^A P_C = \begin{bmatrix} {}^A x_C \\ {}^A y_C \\ {}^A z_C \end{bmatrix}$$

?

$${}^B P_C = \begin{bmatrix} {}^B x_C \\ {}^B y_C \\ {}^B z_C \end{bmatrix}$$



$${}^A P_C = {}^A P_B + \boxed{{}^B P_C} \quad \times$$

$$\cancel{{}^{B}}\ \Rightarrow\ = \ {}^A P_D$$

Rotation Matrix

Euler angles   quaternions

$$\begin{bmatrix}1\\0\\0\end{bmatrix} \qquad \begin{bmatrix}0\\1\\0\end{bmatrix} \qquad \begin{bmatrix}0\\0\\1\end{bmatrix}$$

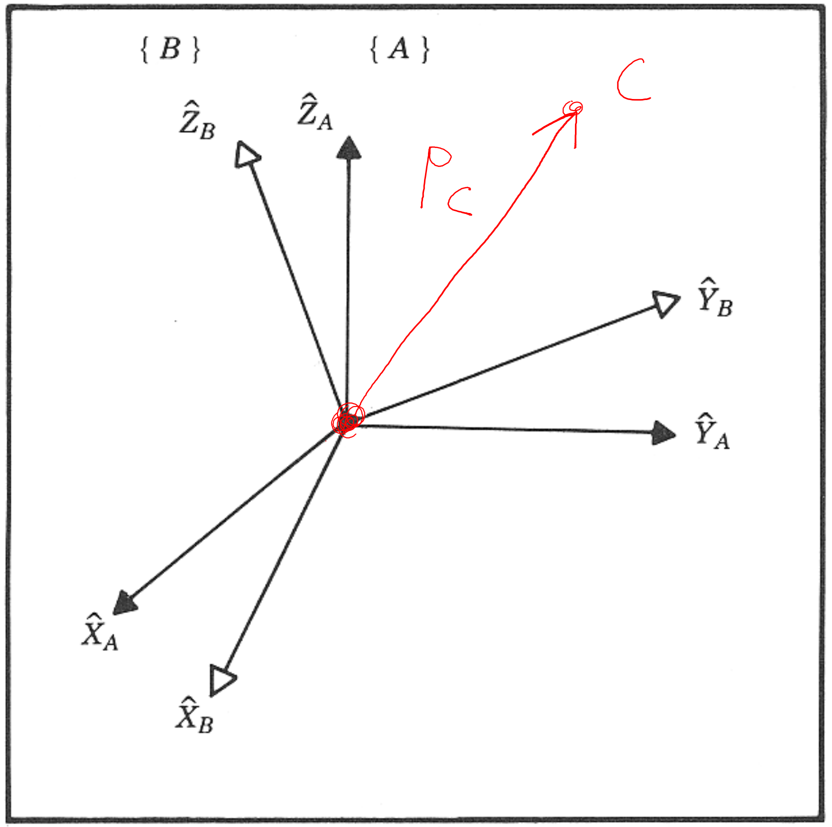
$${}^A p_C \overset{?}{\Longleftrightarrow} {}^B p_C$$

$${}^A p_C = \begin{bmatrix} {}^A x_C \\ {}^A y_C \\ {}^A z_C \end{bmatrix} = {}^A x_C \, {}^A\hat{X}_A + {}^A y_C \, {}^A\hat{Y}_A + {}^A z_C \, {}^A\hat{Z}_A$$

$$= {}^B x_C \, {}^A\hat{X}_B + {}^B y_C \, {}^A\hat{Y}_B + {}^B z_C \, {}^A\hat{Z}_B$$

$$= {}^B x_C \, M \cdot {}^A\hat{X}_A + {}^B y_C \, M \, {}^A\hat{Y}_A + {}^B z_C \, M \, {}^A\hat{Z}_A$$

$$= \boxed{M} \cdot {}^B p_C$$



Rotation Matrix

$R = [x \;\; y \;\; z] \Rightarrow$ ① $\|x\|_2 = x^T x = 1$

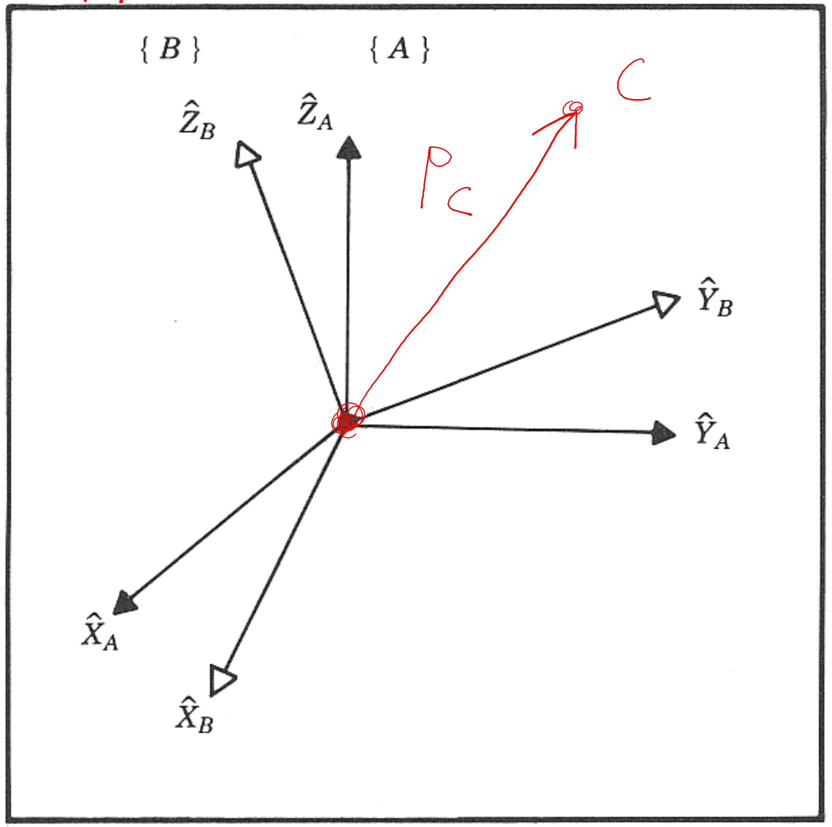
② $x^T y = y^T z = z^T x = 0$

$$\begin{cases} M \cdot {}^A\hat{X}_A = {}^A\hat{X}_B \\ M \cdot {}^A\hat{Y}_A = {}^A\hat{Y}_B \\ M \cdot {}^A\hat{Z}_A = {}^A\hat{Z}_B \end{cases} \Rightarrow M \cdot \underbrace{\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}}_{I} = \underbrace{\begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix}}_{M}$$

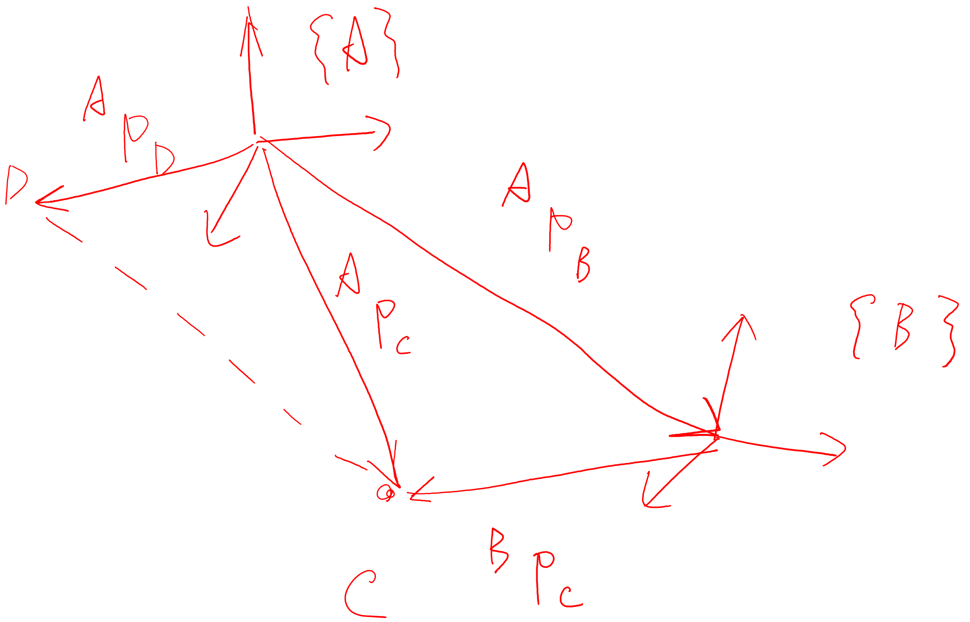
$${}^A p_c = M \, {}^B p_c = {}^A_B R \, {}^B p_c$$

③ $R^T R = \begin{bmatrix} x^T \\ y^T \\ z^T \end{bmatrix} [x \;\; y \;\; z] = \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} = I$

④ $R^T = R^{-1}$



Homogeneous Transformation Matrix



$$\underline{{}^A P_C} = {}^A P_B + {}^A P_D = {}^A P_B + {}^A_B R \, {}^B P_C$$

$${}^A P_D = {}^A_B R \, {}^B P_C$$

$$= \left[{}^A_B R \quad {}^A P_B \right] \begin{bmatrix} {}^B P_C \\ \hline 1 \end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix} {}^A P_C \\ 1 \end{bmatrix}} = \underbrace{\begin{bmatrix} {}^A_B R & {}^A P_B \\ \hline 0 & 1 \end{bmatrix}}_{{}^A_B T} \underbrace{\begin{bmatrix} {}^B P_C \\ 1 \end{bmatrix}}$$

Homogeneous Transformation Matrix

① $${}^{A}_{\cancel{B}}T \cdot {}^{\cancel{B}}_{C}T = {}^{A}_{C}T$$

$$\begin{tikzcd}
\{A\} \arrow[rr, "{}^{A}_{B}T"] \arrow[dr, "{}^{A}_{C}T"'] & & \{B\} \arrow[dl, "{}^{B}_{C}T"] \\
& \{C\} &
\end{tikzcd}$$

$${}^{A}_{\cancel{B}}T \cdot {}^{\cancel{B}}_{A}T = {}^{A}_{A}T = I$$

② $${}^{A}_{B}T \cdot {}^{A}_{B}T^{-1} = I$$

$$\begin{bmatrix} {}^{A}_{B}R & {}^{A}p_{B} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} {}^{A}_{B}R \cdot R & {}^{A}_{B}R \cdot p + {}^{A}p_{B} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \quad R = {}^{A}_{B}R^{-1} = \underline{{}^{A}_{B}R^{T}} \qquad p = -{}^{A}_{B}R^{-1}\,{}^{A}p_{B} = \underline{-{}^{A}_{B}R^{T}\,{}^{A}p_{B}}$$